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MRAM structure with contact plug protruding out of contact hole and method of fabricating the same

Abstract

An MRAM structure includes a dielectric layer. A contact hole is disposed in the dielectric layer. A contact plug fills in the contact hole and protrudes out of the dielectric layer. The contact plug includes a lower portion and an upper portion. The lower portion fills in the contact hole. The upper portion is outside of the contact hole. The upper portion has a top side and a bottom side greater than the top side. The top side and the bottom side are parallel. The bottom side is closer to the contact hole than the top side. An MRAM is disposed on the contact hole and contacts the contact plug.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. application Ser. No. 16/279,956, filed on Feb. 19, 2019. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an MRAM (magnetoresistive random access memory) structure, and more particularly, to a method of fabricating the MRAM structure through an ion beam etch process.

2. Description of the Prior Art

(2) Many modern day electronic devices contain electronic memory configured to store data. Electronic memory may be volatile memory or non-volatile memory. Volatile memory stores data only while it is powered, while non-volatile memory is able to store data when power is removed. MRAM is one promising candidate for next generation non-volatile memory technology. An MRAM cell includes a magnetic tunnel junction (MTJ) having a variable resistance, located between two electrodes disposed within back-end-of-the-line (BEOL) metallization layers.

(3) An MTJ generally includes a layered structure comprising a reference layer, a free layer and a dielectric barrier in between. The reference layer of magnetic material has a magnetic vector that always points in the same direction. The magnetic vector of the free layer is free, but is determined by the physical dimensions of the element. The magnetic vector of the free layer points in either of two directions: parallel or anti-parallel with the magnetization direction of the pinned layer.

(4) Conventional MRAMs have some disadvantages, however. For example, the structures of the contact plugs under the MRAMs have defects. An improved MRAM structure is therefore required in the field.

SUMMARY OF THE INVENTION

(5) According to a preferred embodiment of the present invention, an MRAM structure includes a dielectric layer. A contact hole is disposed in the dielectric layer. A contact plug fills in the contact hole and protrudes out of the dielectric layer, wherein the contact plug comprises a lower portion and an upper portion, the lower portion fills in the contact hole, the upper portion is outside of the contact hole, the upper portion has a top side, a bottom side, a first sloping side and a second sloping side, the top side and the bottom side are parallel, the bottom side is closer to the contact hole than the top side, the bottom side is larger than the top side, two ends of the first sloping side respectively connect the top side and the bottom side, and two ends of the second sloping side respectively connect the top side and the bottom side. An MRAM is disposed on the contact hole

and directly contacts the contact plug.

(6) According to another preferred embodiment of the present invention, a fabricating method of an MRAM structure includes providing a metal line and a dielectric layer covering the metal line. Later, a contact hole in the dielectric layer is formed and the metal line is exposed through the contact hole. Next, a first metal layer is formed to cover and fill the contact hole. Subsequently, a first planarization process is performed to planarize the first metal layer. After that, a bottom electrode, an MTJ material layer and a top electrode are formed to cover the first metal layer after the first planarization process. Finally, an ion beam etch process is performed to pattern the top electrode, the MTJ material layer, the bottom electrode and the first metal layer to form an MRAM and a contact plug.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 to FIG. 8 depict a fabricating method of an MRAM structure according to a first preferred embodiment of the present invention, wherein:

(2) FIG. 1 depicts a dielectric layer with a memory device region and a logic device region;

(3) FIG. 2 is a fabricating stage following FIG. 1;

(4) FIG. 3 is a fabricating stage following FIG. 2;

(5) FIG. 4 is a fabricating stage following FIG. 3;

(6) FIG. 5 is a fabricating stage following FIG. 4;

(7) FIG. 6 is a fabricating stage following FIG. 5;

(8) FIG. 7 is a fabricating stage following FIG. 6; and

(9) FIG. 8 is a fabricating stage following FIG. 7.

(10) FIG. 9 to FIG. 11 depict a fabricating method of an MRAM structure according to a second preferred embodiment of the present invention, wherein:

(11) FIG. 9 depicts a dielectric layer with a memory device region and a logic device region;

(12) FIG. 10 is a fabricating stage following FIG. 9; and

(13) FIG. 11 is a fabricating stage following FIG. 10.

(14) FIG. 12 to FIG. 13 depict a fabricating method of an MRAM structure according to a third preferred embodiment of the present invention, wherein:

(15) FIG. 12 depicts spacers disposed on MRAMs; and

(16) FIG. 13 is a fabricating stage following FIG. 12.

(17) FIG. 14 to FIG. 15 depict a fabricating method of contact plugs according to a fourth preferred embodiment of the present invention, wherein:

(18) FIG. 14 depicts forming another dielectric layer **114** on a dielectric layer; and

(19) FIG. 15 is a fabricating stage following FIG. 14.

DETAILED DESCRIPTION

(20) FIG. 1 to FIG. 8 depict a fabricating method of an MRAM structure according to a first preferred embodiment of the present invention. As shown in FIG. 1, a dielectric layer **10** is provided. The dielectric layer **10** is divided into a memory device region A and a logic device region B. Metal lines **12** are respectively disposed in the dielectric layer **10** within the memory device region A and the logic device region B. In the embodiment of the present invention, three metal lines **12** are disposed in the memory device region A, and one metal line **12** is disposed in the logic device region B. The metal lines **12** may be part of a metal interconnection. The dielectric layer **10** serves as an interlayer dielectric. Next, a dielectric layer **14** is disposed on the dielectric

layer **10**. The dielectric layer **10** and the dielectric layer **14** may be silicon oxide, silicon nitride, silicon carbon nitride, silicon oxynitride or silicon oxycarbonitride. Then, a contact hole **16** is formed in the dielectric layer **14** within the memory device region A to expose the metal lines **12** through the contact hole **16**. There are several contact holes **16** shown in this embodiment. Each of the metal lines **12** respectively serves as a bottom of one contact hole **16**. Later, a barrier **18** is formed to conformally cover the contact holes **16** and the dielectric layer **14**. The barrier **18** is preferably Ti/TiN composite layer, TaN or other suitable conductive materials. Then, a metal layer **20** fills into the contact holes **16** and covers the dielectric layer **14**. The metal layer **20** is preferably tungsten, but not limited thereto. Other metals such as aluminum or copper can be the material of the metal layer **20**. The fabricating step of the barrier **18** and the metal layer **20** may be deposition processes, such as chemical vapor deposition processes, physical vapor deposition processes or atomic layer deposition processes.

(21) As shown in FIG. 2, a planarization process **22** is performed to planarize the metal layer **20**. After planarizing the metal layer **20**, part of the metal layer **20** is still outside of the contact holes **16**, so a top surface of the metal layer **20** is higher than a top surface of the barrier **18**. The planarization process **22** may be a chemical mechanical planarization. As shown in FIG. 3, a bottom electrode **24** is formed to cover the metal layer **20**. Next, the bottom electrode **24** is planarized. According to a preferred embodiment of the present invention, another chemical mechanical planarization can be used to planarize the bottom electrode **24**. The bottom electrode **24** can be tantalum or other metals. The formation of the bottom electrode **24** can be a deposition process such as a chemical vapor deposition process, a physical vapor deposition process or an atomic layer deposition process.

(22) As shown in FIG. 4, a magnetic tunnel junction (MTJ) material layer **26** is formed to cover the bottom electrode **24**. The MTJ material layer **26** can be formed by forming a first ferromagnetic material **28**, an insulating layer **30** and a second ferromagnetic material **32** in sequence. An interlayer **34** can be formed between the insulating layer **30** and the first ferromagnetic material **28**. An interlayer **36** can be formed between the insulating layer **30** and the second ferromagnetic material **32**. The first ferromagnetic material **28** will serve as a reference layer for an MTJ, and a direction of magnetic dipole moment reference layer is fixed. The second ferromagnetic material **32** will serve as a free layer for the MTJ and the free layer alters its direction of magnetic dipole moment based on different circumstances. The insulating layer **30** serves as a tunneling barrier for the MTJ. The materials of the first ferromagnetic material **28** and the second ferromagnetic material **32** can independently select from Co, Pt, Co/Ni alloy, Co/Pd alloy, Fe/B alloy, Co/Pt alloy, Gd/Fe alloy, Co/Fe alloy, Co/Fe/B alloy or Ta/Fe/Co alloy. The insulating layer **30** can be MgO or Al₂O₃. The interlayer **34** and the interlayer **36** can be Co/Fe alloy. Later, a top electrode **38** is formed to cover the second ferromagnetic material **32**. The top electrode **38** can be tantalum or other conductive materials. After that, a mask layer **40** and a patterned photoresist **42** are formed in sequence. The mask layer **40** may include silicon oxide or silicon nitride. The patterned photoresist **42** defines the position of the MRAM structure in the memory device region A. The first ferromagnetic material **28**, the interlayers **34/36**, the insulating layer **30**, the second ferromagnetic material **32**, the top electrode **38** and the mask layer **40** can be formed by deposition processes. The patterned photoresist **42** can be formed by a deposition process and a lithographic process.

(23) As shown in FIG. 5, the mask layer **40** and the top electrode **38** are etched through a reactive ion etch process by using the patterned photoresist **42** as a mask. During the reactive ion etch process, the patterned photoresist **42** may be consumed. After the reactive ion etch process, the mask layer **40** is removed. As shown in FIG. 6, an ion beam etch process **44** is performed to pattern the MTJ material layer **26**, the bottom electrode **24**, the metal layer **20** and the barrier **18** by using the remaining top electrode **38** as a mask. After the ion beam etch process **44**, the MTJ material layer **26**, the top electrode **38** and the bottom electrode are defined as three MRAMs **46a/46b/46c**.

The metal layer **20** below the MRAMs **46a/46b/46c** becomes three contact plugs **48a/48b/48c**.

Each of the MRAMs **46a/46b/46c** includes the top electrode **38**, the MTJ **126** and the bottom electrode **24**. The top electrode **38**, the MTJ material layer **26**, the bottom electrode **24**, the metal layer **20** and the barrier **18** within the logic device region B are removed completely.

(24) Furthermore, the contact plugs **48a/48b/48c** respectively electrically connect to the metal layer **20** and the MRAM **46a/46b/46c**. Each of the contact plugs **48a/48b/48c** respectively includes one of the lower portions **148a/148b/148c** and one of the upper portions **248a/248b/248c**, wherein the lower portions **148a/148b/148c** fill in the corresponding contact holes **16** and the upper portions **248a/248b/248c** are outside of the contact holes **16**. The lower portions **148a/148b/148c** are preferably rectangular. Each of the lower portions **148a/148b/148c** extends from the corresponding contact holes **16** and respectively connects to the corresponding upper portions **248a/248b/248c**. Each of the upper portions **248a/248b/248c** respectively has a top side **52a/52b/52c**, a bottom side **50a/50b/50c**, a first sloping side **54a/54b/54c** and a second sloping side **56a/56b/56c**. The bottom side **50a/50b/50c** is greater than the opening of each of the contact holes **16**. The interface between the upper portions **248a/248b/248c** and the lower portions **148a/148b/148c** are marked by dashed lines. Although the present invention takes three MRAMs **46a/46b/46c** and three contact plugs **48a/48b/48c** as an example, the numbers of MRAMs and contact plugs can be altered based on different requirements.

(25) It is noteworthy that the top electrode **38**, the MTJ material layer **26**, the bottom electrode **24**, the barrier layer **18** and the metal layer **20** are etched by the ion beam etch process **44** rather than a reactive ion etch process. Because the etching ratios of the ion beam etch process **44** to any material are similar values, the MTJ material layer **26**, the top electrode **38** and the bottom electrode **24** can be etched at the same rate during the ion beam etch process **44**. Therefore, the sidewalls of the MRAMs **46a/46b/46c** will become more aligned. On the other hand, as the reactive ion etching process is etched by chemical reactions, the etching ratio to different materials in the reactive ion etching process differs a lot. If the reactive ion etching process is used to etch the top electrode **38**, the MTJ material layer **26**, the bottom electrode **24**, the barrier layer **18** and the metal layer **20**, the sidewall of the MRAMs will become uneven like a stair. This will influence the electric property of the MRAMs **46a/46b/46c**. Moreover, the sidewalls of MRAMs **46a/46b/46c** and the sidewalls contact plugs **48a/48b/48c** formed by the ion beam etch process **44** respectively form several continuous sloping sides, whereas the sidewalls of the MRAMs and the sidewalls of the contact plugs formed by the ion reactive etching process are perpendicular to the top surface of the dielectric layer **10**.

(26) As shown in FIG. 7, a spacer material layer **58** is formed to cover the memory device region A and the logic device region B. The spacer material layer **58** can be silicon nitride. The spacer material layer **58** covers the MRAMs **46a/46b/46c** and the contact plugs **48a/48b/48c**. The first sloping side **54a/54b/54c** and the second sloping side **56a/56b/56c** are entirely covered by the spacer material layer **58**. At this point, the MRAM structure **100** of the present invention is completed. It is noteworthy that the formation of the spacer material layer **58** and the ion beam etch process **44** are performed in the same chamber for preventing the oxidation of metals in the MTJ **126**, avoiding the contaminant formed during the ion beam etch process **44** from being brought to the next chamber and preventing the material layer formed afterwards from contacting the metals in the MTJ **126**.

(27) As shown in FIG. 8, a dielectric layer **60** is formed to cover the spacer material layer **58**. Later, a metal interconnection **62** is formed in the logic region B, wherein the metal interconnection **62** contacts the metal line **12** within the logic region device. B.

(28) FIG. 9 to FIG. 11 depict a fabricating method of an MRAM structure according to a second preferred embodiment of the present invention, wherein elements which are substantially the same as those in the first preferred embodiment are denoted by the same reference numerals; an accompanying explanation is therefore omitted. According to the second preferred embodiment,

after the metal layer **20** is formed as shown in FIG. **1**, the step shown in FIG. **9** is performed. As shown in FIG. **9**, the metal layer **20** and the barrier **18** outside of the contact holes **16** are entirely removed by a planarization process **144**, and the top surface of the dielectric layer **14** is exposed. As shown in FIG. **10**, a metal layer **120** is formed to cover the metal layer **20**, and part of the metal layer **120** directly contacts the top surface of the dielectric layer **14**. As shown in FIG. **11**, a planarization process **244** is performed to planarize the metal layer **120**. After the step shown in FIG. **11**, the steps in FIG. **3** to FIG. **8** can be performed to form the MRAM structure **100**.

(29) FIG. **12** to FIG. **13** depict a fabricating method of an MRAM structure according to a third preferred embodiment of the present invention, wherein elements which are substantially the same as those in the first preferred embodiment are denoted by the same reference numerals; an accompanying explanation is therefore omitted. After forming the spacer material layer **58** as shown in FIG. **12**, the spacer material layer **58** can be etched in the step shown in FIG. **12** to make the spacer material layer **58** become several spacers **158**. The spacers **158** are respectively disposed at two side of each of the MRAMs **46a/46b/46c**. The spacers **158** disposed on different MRAMs are not connected to each other. It is noteworthy that the first sloping sides **54a/54b/54c** and the second sloping sides **56a/56b/56c** of the corresponding contact plugs **48a/48b/48c** are not entirely covered by the spacers **158**. That is, at least part of the spacer material layer **58** on the first sloping sides **54a/54b/54c** and the second sloping sides **56a/56b/56c** are removed. After forming the spacers **58**, the step in FIG. **13** can be performed to form a dielectric layer **60** to cover the spacers **158**. Later, a metal interconnection **62** is formed within the logic device region B. The metal interconnection **62** contacts the metal line **12** in the logic device region B.

(30) As shown in FIG. **7**, an MRAM structure **100** includes a dielectric layer **14**. A contact hole **16** is disposed in the dielectric layer **14**. A contact plug **48b** fills in the contact hole **16** and protrudes out of the dielectric layer **14**. The contact plug **48b** includes a lower portion **148b** and an upper portion **248b**. The lower portion **148b** fills in the contact hole **16**. The upper portion **248b** is outside of the contact hole **16**. The lower portion **148** extends from the contact hole **16** to connect with the upper portion **248b**. The upper portion **248b** has a top side **52b**, a bottom side **50b**, a first sloping side **54b** and a second sloping side **56b**. The top side **52b** and the bottom side **50b** are parallel. The bottom side **50b** is closer to the contact hole **16** than the top side **52b**. The bottom side **50b** is greater than the top side **52b**. Two ends of the first sloping side **54b** respectively connect the top side **52b** and the bottom side **50b**. Two ends of the second sloping side **56b** respectively connect the top side **52b** and the bottom side **50b**. The bottom side **50b** is greater than the opening of the contact hole **16**. A barrier **18** is disposed between the contact plug **48b** and the dielectric layer **14**, and the barrier **18** covers the sidewall of the contact hole **16** and the top surface of the dielectric layer **14**. According to another preferred embodiment of the present invention, the barrier **18** can only cover the inner sidewall of the contact hole **16** and does not cover the top surface of the dielectric layer **14**. The barrier **18** can be Ti/TiN composite layer, tantalum nitride or other suitable conductive materials.

(31) A dielectric layer **10** is disposed below the dielectric layer **14**. A metal line **12** is disposed in the dielectric layer **10** and electrically connects to the contact plug **48b**. The contact plug **48b** is monolithic and is made of a single material. According to a preferred embodiment of the present invention, the contact plug **48b** is preferably made of tungsten, but not limited thereto. Other metals such as aluminum or copper can be used to form the contact plug **48b**.

(32) An MRAM **46b** is disposed on the contact plug **48b** and contacts the contact plug **48b**. The MRAM **46b** includes a MTJ **126**, a top electrode **38** and a bottom electrode **24**. The bottom electrode **24** contacts the contact plug **48b**. The MTJ **126** is between the top electrode **38** and the bottom electrode **24**. The MTJ **126** includes a first ferromagnetic material **28**, an insulating layer **30** and a second ferromagnetic material **32**. The insulating layer **30** is between the first ferromagnetic material **28** and the second ferromagnetic material **32**. An interlayer **34** is between the insulating layer **30** and the first ferromagnetic material **28**. An interlayer **36** is between the insulating layer **30**

and the second ferromagnetic material **32**. A spacer material layer **58** completely covers the first sloping side **54b** and the second sloping side **56b**. The spacer material layer **58** extends to the MRAM **46b**. According to another preferred embodiment of the present invention, as shown in FIG. **12**, a spacer **158** only covers part of the first sloping side **54b** and part of the second sloping side **56b**.

(33) In general, several MRAMs and contact plugs are arranged on the dielectric layer **14** to form a memory array. FIG. **7** takes three MRAMs **46a/46b/46c** and three contact plugs **48a/48b/48c** as an example. Because the MRAMs **46a/46b/46c** and contact plugs **48a/48b/48c** are defined by the ion beam etch process **44**, the contact plug **48b** in the middle has a different profile from that of the contact plugs **48a/48c** at the right side and the left side. The profile of the MRAM **46b** in the middle is also different from that of the MRAMs **46a/46c** at the right side and the left side. Both of the contact plugs **48b** in the middle and the MRAM **46b** in the middle have a left-right symmetric profile. The contact plugs **48a/48c** and the MRAMs **46a/46c** individually have a left-right asymmetric profile. The following description takes the contact plug **48b** in the middle and the contact plug **48c** at the right side as an example. A first angle P1 is disposed between the first sloping side **54b** and the top side **52b** of the contact plug **48b**. A second angle P2 is disposed between the second sloping side **56b** and the top side **54b** of the contact plug **48b**. The size (in degrees) of the first angle P1 equals the size of the second angle P2. A first angle P3 is disposed between the first sloping side **54c** and the top side **52c**. A second angle P4 is disposed between the second sloping side **56c** and the top side **52c**. The size (in degrees) of the first angle P3 does not equal the size of the second angle P4.

(34) FIG. **14** to FIG. **15** depict a fabricating method of contact plugs according to a fourth preferred embodiment of the present invention, wherein elements which are substantially the same as those in the first preferred embodiment are denoted by the same reference numerals; an accompanying explanation is therefore omitted. As shown in FIG. **14**, another dielectric layer **114** is formed on the dielectric layer **14**. Later, a metal layer **20** is formed to cover the dielectric layer **114**. Then, the metal layer **20** is planarized by a chemical mechanical planarization to make the top surface of the metal layer **20** align with the top surface of the dielectric layer **114**. The metal layer **20** which is planarized becomes contact plugs **48d** entirely embedded in the dielectric layer **114**. The disadvantage of the fourth preferred embodiment is that the chemical mechanical planarization leads to dishing on the surface of the dielectric layer **114**, and holes **66** in the contact plugs **48d**. The dishing **64** and holes **66** will influence the property of the MRAM. Therefore, the dielectric layer **114** is omitted in the first preferred embodiment and the second preferred embodiment of the present invention. In this way, the chemical mechanical planarization does not need to stop on the dielectric layer **114** during the planarization of the metal layer **20**, and the dishing **64** and holes **66** can be prevented. Furthermore, the MRAMs **46a/46b/46c** in the first, second and third preferred embodiments of the present invention are formed by the ion beam etch process **44**. Therefore, the sidewalls of the MRAMs **46a/46b/46c** are flat and even.

(35) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A fabricating method of an MRAM structure, comprising: providing a metal line and a dielectric layer covering the metal line; forming a contact hole in the dielectric layer and exposing the metal line through the contact hole; forming a first metal layer covering and filling the contact hole; performing a first planarization process to planarize the first metal layer; forming a bottom electrode, an MTJ material layer and a top electrode covering the first metal layer after the first

planarization process; performing an ion beam etch process to pattern the top electrode, the MTJ material layer, the bottom electrode and the first metal layer at one etching step to form an MRAM and a contact plug, wherein the contact plug is disposed below the MTJ material layer; and forming a spacer material layer covering and physically contacting a sidewall of the top electrode, a sidewall of the MTJ material layer, and a sidewall of the bottom electrode, and covering and contacting the contact plug, wherein the spacer material layer is silicon nitride; wherein the contact plug fills in the contact hole and protrudes from the dielectric layer, the contact plug has a lower portion and an upper portion, the lower portion fills in the contact hole, the upper portion is outside of the contact hole, the upper portion has a top side, a bottom side, a first sloping side and a second sloping side, the top side and the bottom side are parallel, the bottom side is closer to the contact hole than the top side, the bottom side is greater than the top side, two ends of the first sloping side respectively connect the top side and the bottom side, two ends of the second sloping side respectively connect the top side and the bottom side, a first angle is disposed between the first sloping side and the top side, a second angle is disposed between the second sloping side and the top side, and a size of the first angle does not equal to a size of the second angle.

2. The fabricating method of an MRAM structure of claim 1, further comprising etching the spacer material layer to form two spacers at two sides of the MRAM, wherein part of the contact plug is not covered by the spacer.

3. The fabricating method of an MRAM structure of claim 1, wherein the contact plug contacts the metal line.

4. The fabricating method of an MRAM structure of claim 1, wherein after planarizing the first metal layer, part of the first metal layer is outside of the contact hole.

5. The fabricating method of an MRAM structure of claim 1, further comprising: before forming the first metal layer, forming a barrier contacting the contact hole and a top surface of the dielectric layer, wherein after planarizing the first metal layer, a top surface of the first metal layer is aligned with the dielectric layer and the barrier on the top surface of the dielectric layer is removed.

6. The fabricating method of an MRAM structure of claim 5, further comprising: after planarizing the first metal layer, forming a second metal layer covering the first metal layer; and performing a second planarization process to planarize the second metal layer, wherein after planarizing the second metal layer, part of the second metal layer is outside of the contact hole.

7. The fabricating method of an MRAM structure of claim 1, wherein the bottom side is greater than an opening of the contact hole, and under a cross-sectional view, the lower portion is rectangular.

8. The fabricating method of an MRAM structure of claim 1, further comprising: after forming the contact hole in the dielectric layer, forming a barrier covering and contacting the contact hole and the dielectric layer; and after forming the barrier, forming the first metal layer covering and filling the contact hole; wherein the top electrode, the MTJ material layer, the bottom electrode, the first metal layer and the barrier are patterned at one etching step to form the MRAM and the contact plug during the ion beam etch process.
